MH7007 Topics in Scientific Computation I

Project Report: Algorithms, Implementation, and Analysis

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April 16, 2026

Abstract

This report answers Questions 1, 2, and 4 of the project sheet. Question 1 constructs a second-order finite-difference method for a variable-coefficient elliptic boundary value problem and verifies it by a manufactured solution. Question 2 solves a periodic linear dissipation-dispersion equation by Fourier analysis and illustrates the corresponding finite-difference Crank–Nicolson method numerically. Question 4 studies viscous Burgers' equation, including its exact travelling wave, the conservative-form Crank–Nicolson–Leap-Frog method, and a Chebyshev collocation discretisation for the interval problem.

1 Finite Difference Method for a Variable-Coefficient Elliptic Problem

1.1 Taylor expansion and the conservative flux approximation

Let $a(x)$ be smooth and define

$$P(x) = \big(a(x)u'(x)\big)'. \tag{1}$$

The operator is naturally written as the derivative of a flux. Let $h > 0$ be small and introduce the half-grid points $x \pm h/2$. By Taylor expansion,

$$a\left(x + \frac{h}{2}\right) u'\left(x + \frac{h}{2}\right) = a(x)u'(x) + \frac{h}{2}\big(au'\big)'(x) + \frac{h^2}{8}\big(au'\big)''(x) + O(h^3), \tag{2}$$

$$a\left(x - \frac{h}{2}\right) u'\left(x - \frac{h}{2}\right) = a(x)u'(x) - \frac{h}{2}\big(au'\big)'(x) + \frac{h^2}{8}\big(au'\big)''(x) + O(h^3). \tag{3}$$

Subtracting and dividing by h gives

$$P(x) = \big(au'\big)'(x) = \frac{a(x+h/2)u'(x+h/2) - a(x-h/2)u'(x-h/2)}{h} + O(h^2). \tag{4}$$

Now take the uniform grid $x_j = jh$, $j = 0, 1, \ldots, N$, where $h = 1/N$, and let

$$v_j \approx u(x_j), \qquad a_{j+1/2} = a(x_j + h/2). \tag{5}$$

Using centered differences for the half-grid derivatives,

$$u'(x_{j+1/2}) \approx \frac{v_{j+1} - v_j}{h}, \qquad u'(x_{j-1/2}) \approx \frac{v_j - v_{j-1}}{h}. \tag{6}$$

Therefore

$$P(x_j) \approx \frac{a_{j+1/2}v_{j+1} - (a_{j+1/2} + a_{j-1/2})v_j + a_{j-1/2}v_{j-1}}{h^2}. \tag{7}$$

This is second-order accurate for smooth a and u.

1.2 Discrete scheme for the boundary value problem

We solve

$$-\big(a(x)u'(x)\big)' + c(x)u(x) = f(x), \qquad 0 < x < 1, \qquad u(0) = u(1) = 0, \tag{8}$$

where $a(x) \geq a_0 > 0$. At the interior nodes x_j, $j = 1, \ldots, N-1$, substituting (7) into (8) gives

$$-\frac{a_{j-1/2}}{h^2} v_{j-1} + \left(\frac{a_{j+1/2} + a_{j-1/2}}{h^2} + c_j \right) v_j - \frac{a_{j+1/2}}{h^2} v_{j+1} = f_j, \tag{9}$$

where $c_j = c(x_j)$ and $f_j = f(x_j)$. Since $v_0 = v_N = 0$, the unknown vector is

$$V = (v_1, v_2, \ldots, v_{N-1})^T. \tag{10}$$

The linear system is tridiagonal:

$$AV = F, \tag{11}$$

with diagonal entries

$$A_{j,j} = \frac{a_{j+1/2} + a_{j-1/2}}{h^2} + c_j, \tag{12}$$

upper diagonal entries

$$A_{j,j+1} = -\frac{a_{j+1/2}}{h^2}, \tag{13}$$

and lower diagonal entries

$$A_{j,j-1} = -\frac{a_{j-1/2}}{h^2}. \tag{14}$$

This is the scheme implemented in the notebook using banded storage and a tridiagonal solver.

1.3 Manufactured solution test

For verification, choose

$$u(x) = x \sin(2\pi x), \qquad a(x) = 1 + x^2, \qquad c(x) = e^x. \tag{15}$$

This exact solution satisfies $u(0) = u(1) = 0$. We manufacture the forcing term by substituting this u into the differential equation:

$$f(x) = -(a(x)u'(x))' + c(x)u(x). \tag{16}$$

The required derivatives are

$$u'(x) = \sin(2\pi x) + 2\pi x \cos(2\pi x), \tag{17}$$

and

$$u''(x) = 4\pi \cos(2\pi x) - 4\pi^2 x \sin(2\pi x). \tag{18}$$

Since $a'(x) = 2x$,

$$(au')' = a'(x)u'(x) + a(x)u''(x), \tag{19}$$

so the manufactured right-hand side is

$$\begin{aligned} f(x) = &-2x\,(\sin(2\pi x) + 2\pi x \cos(2\pi x)) \\ &- (1 + x^2)\,\big(4\pi \cos(2\pi x) - 4\pi^2 x \sin(2\pi x)\big) + e^x x \sin(2\pi x). \end{aligned} \tag{20}$$

Table 1: Maximum-norm error for the manufactured solution. The observed rates are close to 2, confirming the expected second-order accuracy.

N	h	$\|e_h\|_\infty$	observed rate
10	0.10000	2.911×10^{-2}	–
20	0.05000	7.113×10^{-3}	2.033
40	0.02500	1.768×10^{-3}	2.008
80	0.01250	4.421×10^{-4}	2.000
160	0.00625	1.105×10^{-4}	2.001
320	0.00313	2.762×10^{-5}	2.000

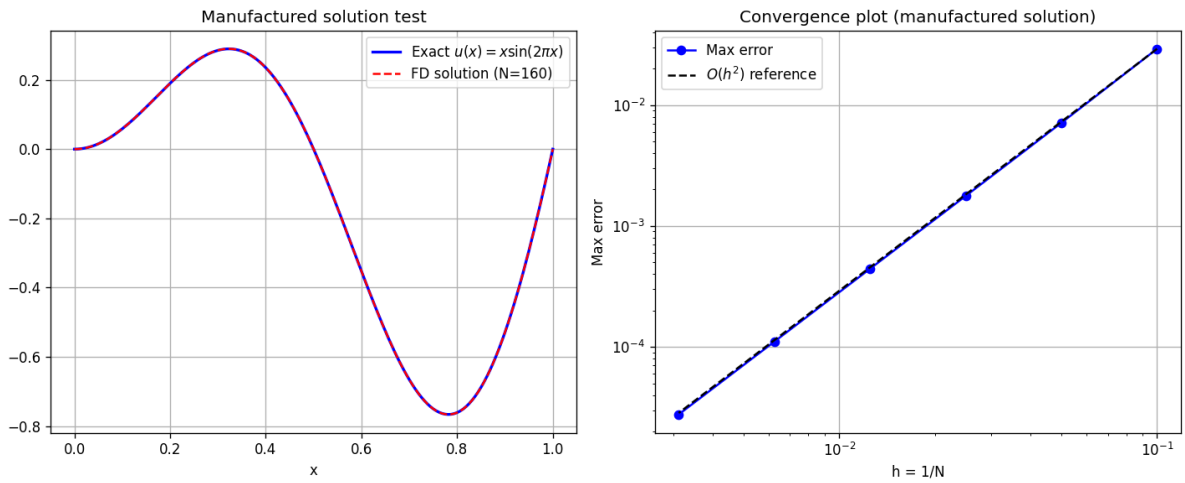


Figure 1: Manufactured solution test. The finite-difference solution for $N = 160$ is visually indistinguishable from the exact solution, and the log-log error plot follows the $O(h^2)$ reference line.

1.4 Numerical solution for $f(x) = 1$

Finally, use the same coefficients

$$a(x) = 1 + x^2, \qquad c(x) = e^x, \tag{21}$$

but set $f(x) = 1$. The problem becomes

$$-\big((1 + x^2)u'(x)\big)' + e^x u(x) = 1, \qquad 0 < x < 1, \qquad u(0) = u(1) = 0. \tag{22}$$

The numerical computation uses $N = 400$. Since $a(x) > 0$, $c(x) > 0$, and $f(x) > 0$, the maximum principle suggests a nonnegative interior solution. The computed solution satisfies

$$\min_j v_j \approx 0, \qquad \max_j v_j \approx 0.083874. \tag{23}$$

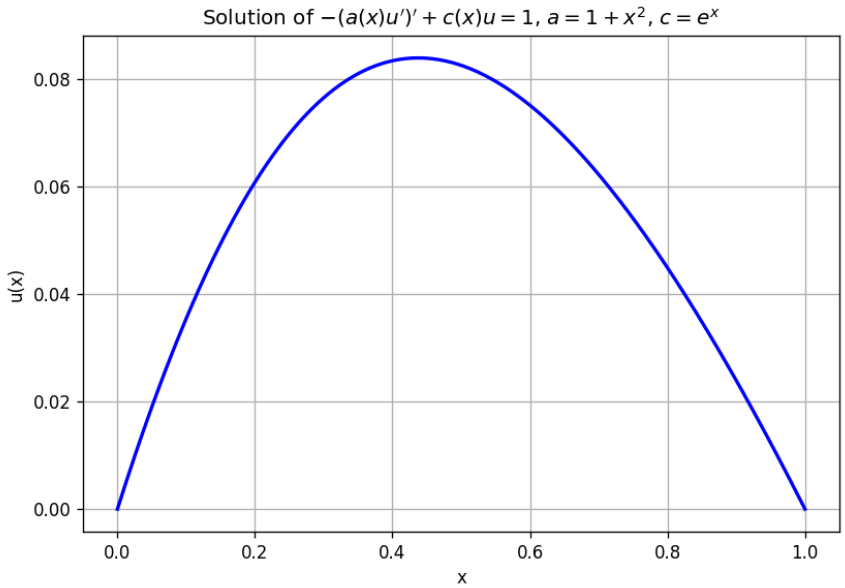


Figure 2: Finite-difference solution of (22). The solution is nonnegative, vanishes at both boundaries, and reaches its maximum in the interior.

2 Dissipation and Dispersion

2.1 Fourier solution of the periodic PDE

Consider

$$u_t + au_x = \alpha u_{xx} - \beta u_{xxx}, \qquad -\pi < x < \pi, \quad t > 0, \tag{24}$$

with 2π-periodic boundary conditions and initial data

$$u(x, 0) = u_0(x). \tag{25}$$

Assume u_0 is smooth and 2π-periodic. Seek a Fourier series solution

$$u(x, t) = \sum_{k=-\infty}^{\infty} v_k(t) e^{ikx}. \tag{26}$$

For one Fourier mode,

$$\partial_x e^{ikx} = ike^{ikx}, \qquad \partial_{xx} e^{ikx} = -k^2 e^{ikx}, \qquad \partial_{xxx} e^{ikx} = -ik^3 e^{ikx}. \tag{27}$$

Substituting (26) into (24) gives the decoupled ODE

$$v_k'(t) = \left(-iak - \alpha k^2 + i\beta k^3\right) v_k(t). \tag{28}$$

Thus

$$v_k(t) = v_k(0) \exp\left\{\left(-iak - \alpha k^2 + i\beta k^3\right) t\right\}. \tag{29}$$

The full solution is therefore

$$u(x, t) = \sum_{k=-\infty}^{\infty} v_k(0) \exp(-\alpha k^2 t) \exp\left(i\left[kx - (ak - \beta k^3)t\right]\right). \tag{30}$$

2.2 Dissipation versus dispersion

Equation (30) separates amplitude and phase effects. The factor

$$\exp(-\alpha k^2 t) \tag{31}$$

controls amplitude. If $\alpha > 0$, then all nonzero modes decay, and higher-frequency modes decay faster because the decay rate is proportional to k^2. This is dissipation.

The oscillatory phase has angular frequency

$$\omega(k) = ak - \beta k^3. \tag{32}$$

The phase speed is

$$c_p(k) = \frac{\omega(k)}{k} = a - \beta k^2. \tag{33}$$

If $\beta \neq 0$, then different wave numbers travel at different speeds. This frequency dependence changes the wave shape over time and is called dispersion.

2.3 Second-order finite-difference Crank–Nicolson scheme

On a periodic grid

$$x_j = -\pi + jh, \qquad h = \frac{2\pi}{N}, \qquad j = 0, 1, \ldots, N-1, \tag{34}$$

use centered finite differences in space. For the discrete Fourier mode e^{ikx_j}, the second-order symbols are

$$\lambda_1(k) = \frac{i\sin(kh)}{h}, \tag{35}$$

$$\lambda_2(k) = \frac{2(\cos(kh) - 1)}{h^2}, \tag{36}$$

$$\lambda_3(k) = \frac{i(\sin(2kh) - 2\sin(kh))}{h^3}. \tag{37}$$

These approximate the continuous symbols ik, $-k^2$, and $-ik^3$ with $O(h^2)$ error. Hence the discrete spatial symbol is

$$L_k^h = -a\lambda_1(k) + \alpha\lambda_2(k) - \beta\lambda_3(k). \tag{38}$$

Applying Crank–Nicolson in time to $U_k'(t) = L_k^h U_k(t)$ gives

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} = \frac{1}{2} L_k^h \left(U_k^{n+1} + U_k^n\right), \tag{39}$$

so the Fourier-mode amplification factor is

$$G_k = \frac{1 + \frac{\Delta t}{2} L_k^h}{1 - \frac{\Delta t}{2} L_k^h}. \tag{40}$$

This is second order in time, while the centered spatial discretization is second order in space.

2.4 Numerical illustrations

The numerical experiments use the smooth periodic initial condition

$$u_0(x) = \sin x + 0.6\cos(2x) + 0.3\sin(3x), \tag{41}$$

with $a = 1$. Figure 3 compares pure advection, dissipation, and dispersion. In the dissipative case the solution is smoothed; in the dispersive case the profile changes because the component waves propagate at different speeds.

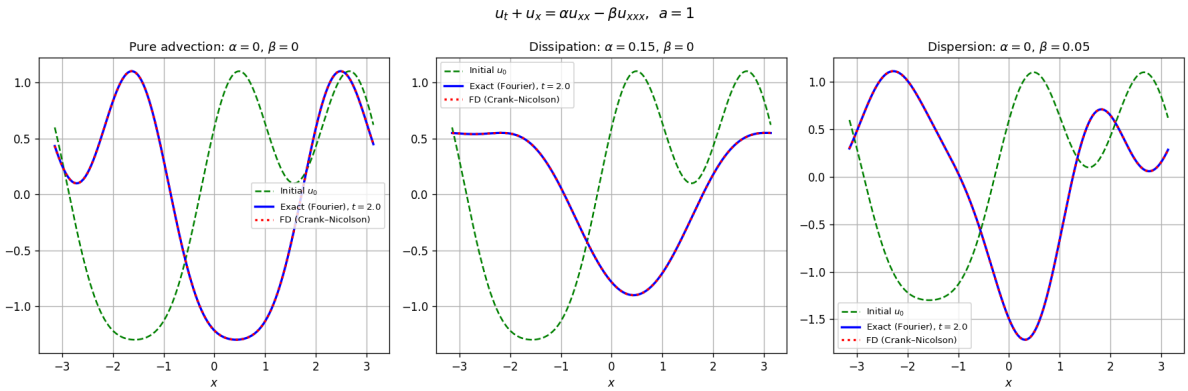


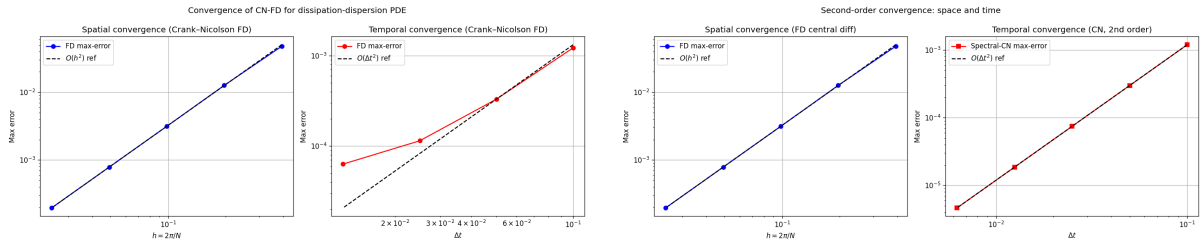
Figure 3: Numerical illustration of pure advection, dissipation, and dispersion for the periodic linear equation. The finite-difference Crank–Nicolson solution agrees closely with the exact Fourier solution.

Table 2: Spatial convergence for the Crank–Nicolson finite-difference method with a very small time step. The observed rates approach 2, as expected for centered second-order spatial differences.

N	$h = 2\pi/N$	max error	observed rate
16	0.3927	4.765×10^{-2}	–
32	0.1963	1.263×10^{-2}	1.916
64	0.0982	3.136×10^{-3}	2.009
128	0.0491	7.849×10^{-4}	1.999
256	0.0245	1.963×10^{-4}	2.000

Table 3: Temporal convergence for Crank–Nicolson using the exact spectral spatial operator. Removing the finite-difference spatial error exposes the clean second-order time convergence.

Δt	max error	observed rate
0.10000	1.189×10^{-3}	–
0.05000	2.978×10^{-4}	1.997
0.02500	7.447×10^{-5}	1.999
0.01250	1.862×10^{-5}	2.000
0.00625	4.655×10^{-6}	2.000



(a) Finite-difference spatial and temporal convergence.

(b) Spatial convergence and spectral-in-space temporal convergence.

Figure 4: Convergence plots for the second-order methods. The reference slopes confirm the expected $O(h^2)$ and $O(\Delta t^2)$ behavior.

3 Numerical Study of the Burgers' Equation

Consider the viscous Burgers' equation

$$u_t + uu_x = \varepsilon u_{xx}, \qquad x \in \mathbb{R}, \quad t > 0, \qquad \varepsilon > 0. \tag{42}$$

It is useful to write the nonlinear transport term in conservative form:

$$u_t + \frac{1}{2}(u^2)_x = \varepsilon u_{xx}. \tag{43}$$

This form is the natural one for the finite-difference method below, because the nonlinear term is approximated as a centred difference of the flux $u^2/2$.

3.1 Exact travelling-wave solution

We first show that (42) has the exact travelling wave

$$u(x,t) = \omega\left(1 - \tanh\left[\frac{\omega}{2\varepsilon}(x - \omega t - x_0)\right]\right), \tag{44}$$

where ω and x_0 are constants. Set

$$y = x - \omega t - x_0, \qquad u(x,t) = w(y). \tag{45}$$

Then

$$u_t = -\omega w', \qquad u_x = w', \qquad u_{xx} = w''. \tag{46}$$

Substitution into (42) gives

$$-\omega w' + ww' = \varepsilon w'', \tag{47}$$

or, equivalently,

$$-\omega w' + \left(\frac{w^2}{2}\right)' = \varepsilon w''. \tag{48}$$

Integrating once with respect to y,

$$-\omega w + \frac{w^2}{2} = \varepsilon w' + C. \tag{49}$$

For the wave connecting $w(-\infty) = 2\omega$ to $w(+\infty) = 0$, with $w'(\pm\infty) = 0$, the constant is $C = 0$. Hence

$$\varepsilon w' = -\omega w + \frac{w^2}{2} = \frac{w(w - 2\omega)}{2}. \tag{50}$$

This first-order ODE is separable:

$$\frac{dw}{w(w-2\omega)} = \frac{dy}{2\varepsilon}. \tag{51}$$

Using partial fractions,

$$\frac{1}{2\omega}\int\left(\frac{1}{w-2\omega} - \frac{1}{w}\right) dw = \frac{y}{2\varepsilon}. \tag{52}$$

After absorbing the integration constant into the centre x_0, this gives

$$w(y) = \frac{2\omega}{1+\exp(\omega y/\varepsilon)}. \tag{53}$$

Using

$$\frac{2}{1+e^A} = 1 - \tanh\left(\frac{A}{2}\right), \tag{54}$$

with $A = \omega y/\varepsilon$, we obtain exactly (44).

For a direct substitution check, set

$$\eta = \frac{\omega}{2\varepsilon}(x - \omega t - x_0), \qquad u = \omega(1 - \tanh\eta). \tag{55}$$

Since $\eta_x = \omega/(2\varepsilon)$ and $\eta_t = -\omega^2/(2\varepsilon)$,

$$u_x = -\frac{\omega^2}{2\varepsilon}\operatorname{sech}^2\eta, \tag{56}$$

$$u_t = \frac{\omega^3}{2\varepsilon}\operatorname{sech}^2\eta, \tag{57}$$

$$u_{xx} = \frac{\omega^3}{2\varepsilon^2}\operatorname{sech}^2\eta\,\tanh\eta. \tag{58}$$

Therefore

$$u_t + uu_x = \frac{\omega^3}{2\varepsilon}\operatorname{sech}^2\eta\,\tanh\eta = \varepsilon u_{xx}. \tag{59}$$

Thus (44) satisfies the viscous Burgers' equation.

3.2 Crank–Nicolson–Leap-Frog scheme

Let

$$x_m = x_L + mh, \qquad t_n = nk, \qquad V_m^n \approx u(x_m, t_n). \tag{60}$$

From the conservative form (43), approximate the time derivative at (x_m, t_n) by the leap-frog difference

$$u_t(x_m, t_n) \approx \frac{V_m^{n+1} - V_m^{n-1}}{2k}. \tag{61}$$

The nonlinear flux is treated explicitly at time level n:

$$\frac{1}{2}(u^2)_x(x_m, t_n) \approx \frac{1}{2}\frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{2h} = \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h}. \tag{62}$$

The diffusion term is averaged by Crank–Nicolson over the two leap-frog levels:

$$u_{xx}(x_m, t_n) \approx \frac{1}{2h^2}\left(\delta_x^2 V_m^{n+1} + \delta_x^2 V_m^{n-1}\right), \tag{63}$$

where

$$\delta_x^2 V_m^n = V_{m+1}^n - 2V_m^n + V_{m-1}^n. \tag{64}$$

The CNLF scheme is therefore

$$\frac{V_m^{n+1} - V_m^{n-1}}{2k} + \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h} = \frac{\varepsilon}{2h^2}\left(\delta_x^2 V_m^{n+1} + \delta_x^2 V_m^{n-1}\right). \tag{65}$$

Multiplying by $2k$ and setting

$$\mu = \frac{\varepsilon k}{h^2}, \tag{66}$$

we obtain the linear tridiagonal solve at each time step:

$$V_m^{n+1} - \mu\delta_x^2 V_m^{n+1} = V_m^{n-1} + \mu\delta_x^2 V_m^{n-1} - \frac{k}{2h}\left((V_{m+1}^n)^2 - (V_{m-1}^n)^2\right). \tag{67}$$

The nonlinear term is explicit, while the viscous part is implicit. Consequently each update requires only a tridiagonal linear solve. Since the method is two-step, the computation needs V^0 and V^1. In the travelling-wave convergence tests, V^1 is taken from the exact solution to isolate the CNLF discretisation error.

3.3 Local truncation error

Let $u_m^n = u(x_m, t_n)$ be the exact smooth solution sampled on the grid and let $q = u^2$. The centred time difference satisfies

$$\frac{u_m^{n+1} - u_m^{n-1}}{2k} = u_t + \frac{k^2}{6}u_{ttt} + O(k^4). \tag{68}$$

For the nonlinear conservative flux,

$$\frac{(u_{m+1}^n)^2 - (u_{m-1}^n)^2}{4h} = \frac{1}{2}\frac{q(x_m + h, t_n) - q(x_m - h, t_n)}{2h} = \frac{1}{2}q_x + \frac{h^2}{12}q_{xxx} + O(h^4). \tag{69}$$

For the Crank–Nicolson diffusion average,

$$\frac{1}{2h^2}\left(\delta_x^2 u_m^{n+1} + \delta_x^2 u_m^{n-1}\right) = u_{xx} + \frac{k^2}{2}u_{xxtt} + \frac{h^2}{12}u_{xxxx} + O(k^4 + k^2h^2 + h^4). \tag{70}$$

Substituting the exact solution into the discrete scheme and using (43), the local truncation error has leading terms

$$\tau_m^n = \frac{k^2}{6}u_{ttt} + \frac{h^2}{12}(u^2)_{xxx} - \varepsilon\left(\frac{k^2}{2}u_{xxtt} + \frac{h^2}{12}u_{xxxx}\right) + O(k^4 + k^2h^2 + h^4). \tag{71}$$

Therefore

$$\tau_m^n = O(k^2 + h^2). \tag{72}$$

With a second-order startup and stability on the smooth branch of the solution, the expected global error is also second order in space and time.

3.4 Travelling-wave numerical test and convergence

For the verification experiment, use

$$\omega = 0.5, \qquad \varepsilon = 0.1, \qquad x_0 = 3, \qquad -5 \le x \le 5, \qquad T = 2. \tag{73}$$

With $h = k = 10^{-3}$, the notebook computation gives

$$\|e(T)\|_\infty = 4.548 \times 10^{-7}. \tag{74}$$

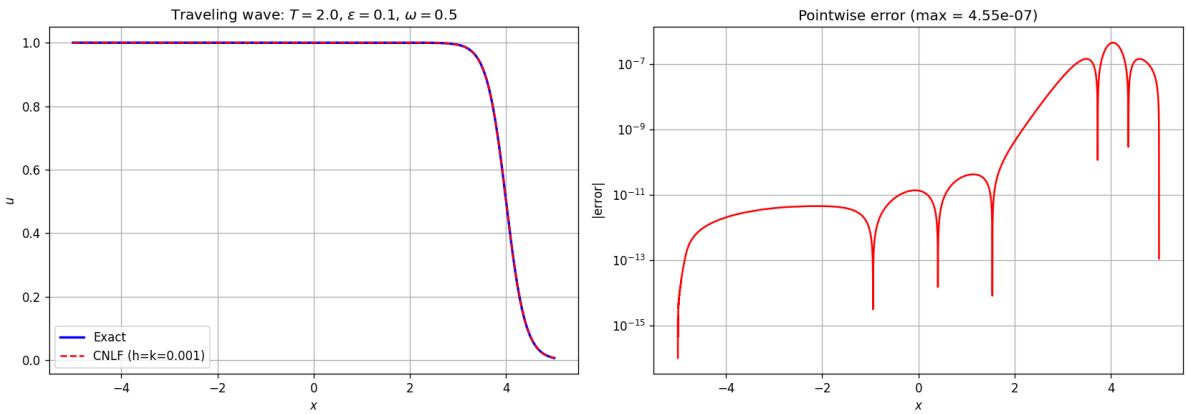


Figure 5: CNLF approximation of the exact Burgers travelling wave at $T = 2$, together with the pointwise absolute error.

Table 4: Coupled refinement study for the travelling-wave solution. The slope is essentially second order when $h = k$.

$h = k$	$\|e(T)\|_\infty$	observed rate
0.0400	6.056×10^{-4}	–
0.0200	1.513×10^{-4}	2.001
0.0100	3.790×10^{-5}	1.997
0.0050	9.485×10^{-6}	1.998

Table 5: Separate temporal convergence with $h = 2.5 \times 10^{-4}$.

k	$\|e\|_\infty$	$\|e\|_2$	L^2 rate
8.0×10^{-3}	1.906×10^{-5}	3.663×10^{-6}	–
4.0×10^{-3}	4.742×10^{-6}	9.125×10^{-7}	2.005
2.0×10^{-3}	1.165×10^{-6}	2.250×10^{-7}	2.020
1.0×10^{-3}	2.708×10^{-7}	5.316×10^{-8}	2.081

Table 6: Separate spatial convergence with $k = 10^{-4}$.

h	$\|e\|_\infty$	$\|e\|_2$	L^2 rate
4.0×10^{-2}	8.461×10^{-4}	1.584×10^{-4}	–
2.0×10^{-2}	2.117×10^{-4}	3.963×10^{-5}	1.999
1.0×10^{-2}	5.295×10^{-5}	9.911×10^{-6}	1.999
5.0×10^{-3}	1.324×10^{-5}	2.478×10^{-6}	2.000

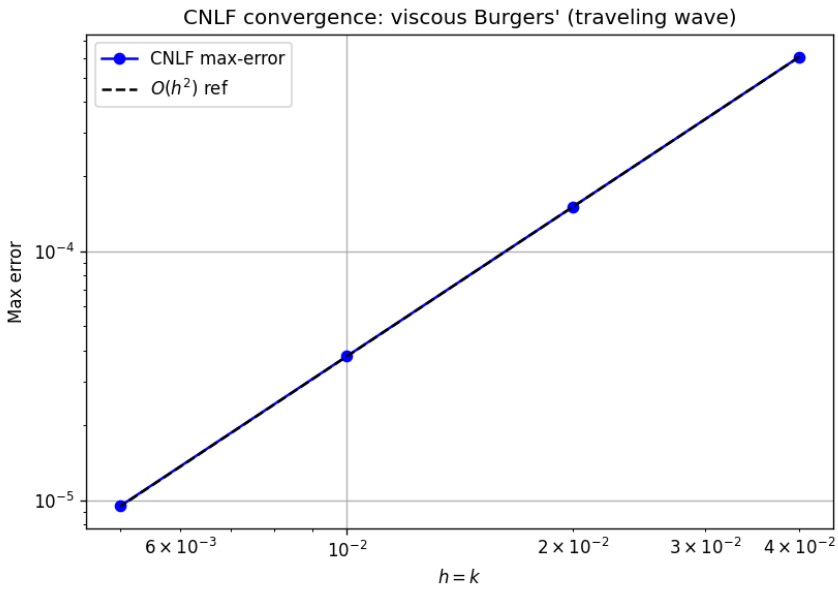


Figure 6: Coupled second-order convergence for the CNLF method on the travelling-wave problem.

3.5 Chebyshev collocation for the interval problem

The last part considers Burgers' equation on $(-1, 1)$ with

$$u(\pm 1, t) = 0, \qquad u(x, 0) = -\sin(\pi x), \qquad \varepsilon = 0.02, \qquad T = 0.995. \tag{75}$$

Fourier collocation is the standard spectral collocation method for periodic domains, because trigonometric basis functions automatically respect periodicity. On a bounded nonperiodic interval, the analogous standard spectral collocation method uses Chebyshev polynomials and Chebyshev–Lobatto points. These nodes cluster near the endpoints, which is useful for viscous Burgers' equation when steep gradients develop near boundaries.

For polynomial degree N, define

$$x_j = \cos\left(\frac{j\pi}{N}\right), \qquad j = 0, 1, \ldots, N. \tag{76}$$

Let $U_j(t) \approx u(x_j, t)$. If p_N is the interpolating polynomial through the nodal values, then differentiation at the nodes is represented by a dense matrix D:

$$p_N'(x_i) \approx \sum_{j=0}^{N} D_{ij} U_j. \tag{77}$$

For $i \neq j$,

$$D_{ij} = \frac{c_i}{c_j} \frac{(-1)^{i+j}}{x_i - x_j}, \qquad c_0 = c_N = 2, \quad c_j = 1 \quad (1 \leq j \leq N-1), \tag{78}$$

and the diagonal is chosen so each row sums to zero:

$$D_{ii} = -\sum_{j \neq i} D_{ij}. \tag{79}$$

The second derivative is represented by

$$D^{(2)} = D^2. \tag{80}$$

Using the conservative form, the semi-discrete collocation system is

$$\dot{U} + \frac{1}{2} D(U \odot U) = \varepsilon D^2 U, \tag{81}$$

where $\odot$ denotes componentwise multiplication. Imposing $U_0 = U_N = 0$ and writing I for the interior indices, the CNLF update becomes

$$\left(I - \varepsilon k D_{II}^2\right) U_I^{n+1} = U_I^{n-1} + \varepsilon k (D^2 U^{n-1})_I - \frac{k}{2} \left(D\left((U^n)^2\right)\right)_I . \tag{82}$$

As before, the method is two-step. The notebook uses one nonlinear Crank–Nicolson startup step:

$$U^1 - U^0 + \frac{k}{4} D\left((U^1)^2 + (U^0)^2\right) - \frac{\varepsilon k}{2} D^2 (U^1 + U^0) = 0, \tag{83}$$

restricted to interior nodes and solved by Newton's method.

For the computation, the notebook uses

$$N = 96, \qquad k = 10^{-3}, \qquad \varepsilon = 0.02, \qquad T = 0.995. \tag{84}$$

Figure 7 shows the resulting Chebyshev-collocation CNLF solution and selected profiles.

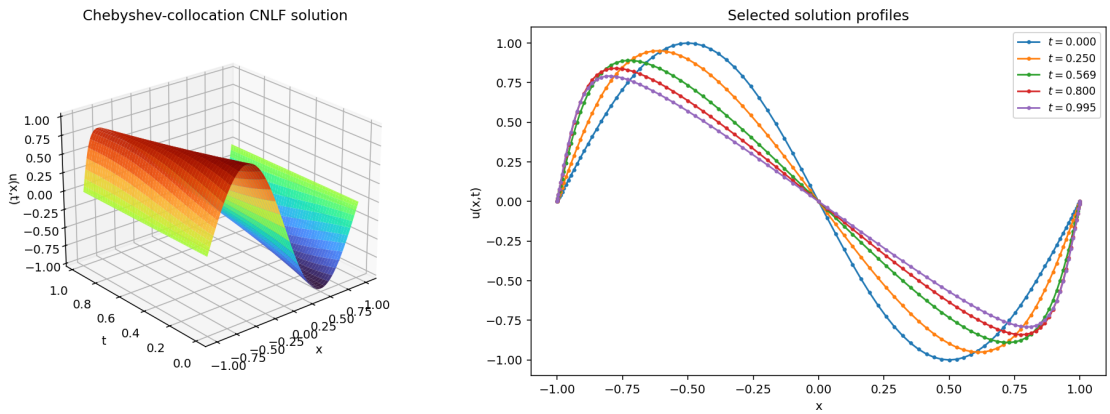


Figure 7: Chebyshev-collocation CNLF solution of Burgers' equation on $(-1, 1)$ with $u(\pm 1, t) = 0$ and $u(x, 0) = -\sin(\pi x)$.

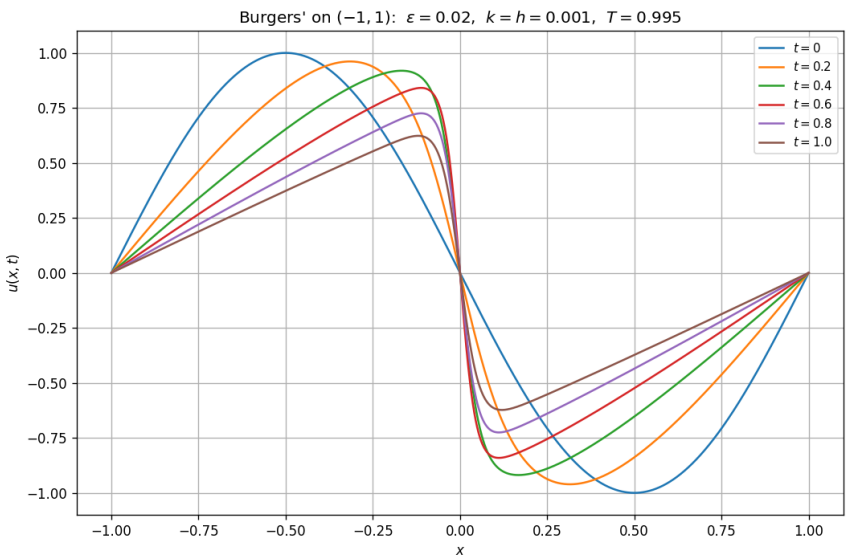


Figure 8: Finite-difference CNLF reference profiles for the same interval problem. The Chebyshev collocation method captures the same steepening and smoothing behaviour using a spectral spatial discretisation.

4 Conclusion

For Question 1, the conservative half-grid flux discretization gives a tridiagonal second-order finite-difference method for the variable-coefficient elliptic problem. The manufactured solution test confirms the $O(h^2)$ convergence rate, and the $f(x) = 1$ computation produces a physically consistent nonnegative solution.

For Question 2, Fourier analysis shows precisely how the parameter α produces dissipation through amplitude decay and how β produces dispersion through frequency-dependent phase speed. The periodic Crank–Nicolson finite-difference method reproduces these effects and demonstrates second-order accuracy in space and time.

For Question 4, the viscous Burgers' equation admits an exact tanh travelling wave, obtained either by travelling-wave reduction or by direct substitution. The CNLF method follows cleanly from the conservative form of Burgers' equation by combining leap-frog time differencing, explicit centred flux differencing, and Crank–Nicolson averaging of the viscous term. Taylor expansion gives local truncation error $O(k^2 + h^2)$, and the numerical tests confirm second-order behaviour. On the nonperiodic interval $(-1, 1)$, Chebyshev collocation provides the corresponding spectral spatial discretisation and resolves the steep solution profiles with endpoint-clustered nodes.